

e/m Experiment

1. Introduction

In this laboratory we are going to carry out two classic experiments at the heart of modern physics: measuring the e/m ratio (the ratio of the electron charge to its mass), and the photoelectric effect. Though relatively simple to perform, both experiments convey a large amount of data about the quantum nature of the world—one points to the quantization of charge, and the other to the quantization of light energy.

As a simple review, recall that moving charges in a magnetic field will feel a force that can cause them to turn in a circular path. The force felt by the charges (here assumed to be elementary charges e) come from the Lorentz equation, and can be set equal to the centripetal force. Then

$$ev \times B = evB = \frac{mv^2}{R} \Rightarrow \frac{e}{m} = \frac{v}{RB}$$

Now we have to get the velocity. This is easy because we can control the speed of the electrons using an accelerating voltage. We then get

$$\frac{1}{2}mv^2 = eV \Rightarrow \frac{e}{m} = \frac{2V}{B^2 R^2}$$

The voltage is known from the readout on the power supply. Onto the second problem. How do we get the magnetic field? Well, we will be using Helmholtz coils to create the field, so there is an equation for the field magnitude. It is

$$B = \frac{8\mu_0 NI}{\sqrt{125}a} = 1.96 \times 10^{-4} I$$

$$\frac{e}{m} = \left(\frac{3.91a^2}{\mu_o^2 N^2} \right) \left(\frac{V}{I^2 r^2} \right).$$

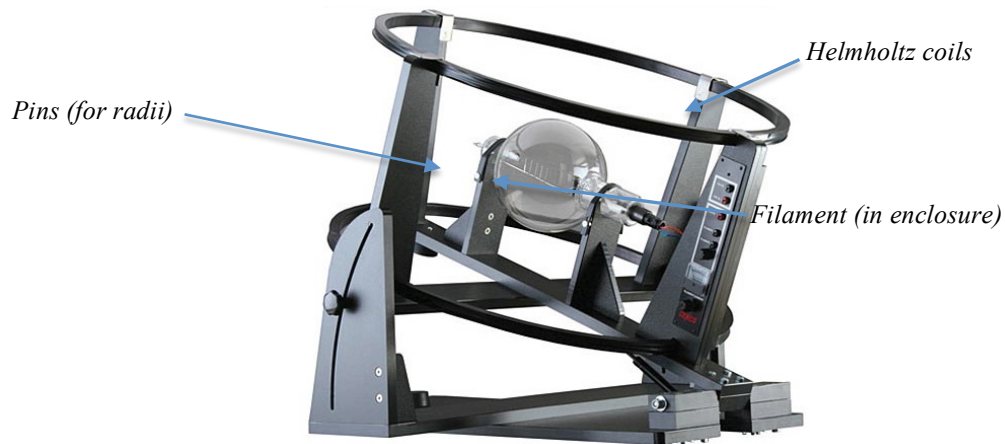
where we used the facts that $N = 72$, and $a=0.33$ for our apparatus, to simplify this working equation. So here is the essence of the experiment: for a particular accelerating voltage (V) we will apply a variety of Helmholtz coil currents to bend the beam. By measuring these currents and the respective radii (R) for the beam, we can calculate the magnetic field (B) and then, finally, get the e/m ratio.

WARNINGS

- Please **DO NOT** touch the filament power supply knobs !
- The filament current has to be kept below 4.5A !
- Before pushing the ON button of Accelerating and Coil power supplies, turn both voltage and current knobs to zero.

2. Setup

Typically the instructor will have made sure that the apparatus is setup. The image below shows the key components.



- The filament of the e/m tube has the proper electron emission when carrying 3.8 – 4.5 A current.
- Good results can be obtained using an accelerating voltage of 23 to 45 volts.
- An electron emission current of about 2 to 4 mA gives a visible beam.
- For maximum filament life operate the tube with the minimum filament current necessary for proper electron emission.

3- Experiment

1. Turn on the filament power supply. Your instructor will carefully increase the current to 3.8 – 4 A. Then darken the room. The filament will take a few minutes to heat up. First you will see and

orange glow through the slit and then you should see the blue beam of electrons.

2. Turn on the accelerating potential power supply. Increase the voltage to 25 V.
3. In the absence of magnetic and electric fields, the electron will travel in a straight path. The Earth's field, however, will bend the beam toward the back of the tube. Turn on the Helmholtz coils power supply and turn up the current until the beam is straight. You can tell the beam is straight when the electron beam is in the center of the light streak that the filament shines on the side of the tube. Record the current in the coils. You will need this current to correct your measurements of B . Estimate the uncertainty in your measurement of the current.
4. Continue to increase the coil current until the outside of the electron beam hits the outside of the outermost peg. The best viewing position is directly above the tube. Record the filament current, the magnet current, the beam current, and the accelerating voltage. Repeat the procedure for the inner pegs. Note that in some cases you may not be able to "hit" each and every crossbar. Take as many as you can.
5. Repeat the determination of e/m for total 3 accelerating potentials between 20 V and 40 V.
6. Finishing up. Turn the filament current down to zero. Turn down the accelerating voltage. Turn off all power supplies, and disconnect the electron tube from the other circuitry.

4- Analysis

- For each crossbar, $R = D/2$, where the values for D are given below.

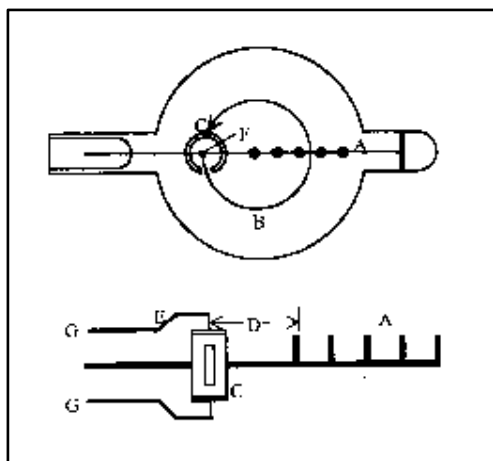


Figure (a) A top view of the e/m tube.
 (b) A crosssectional view of the filament assembly.

Crossbar number	Distance to Filament ($D=2R$)
1	0.065 m
2	0.078 m
3	0.090 m
4	0.103 m
5	0.115 m

1. The vertical component of the Earth's magnetic field is approximately 5.2×10^{-5} T. Part of the field generated by the coils is used to counteract this field. Since the field of the coils is proportional to the current, subtract the portion of the current needed to cancel the field of the Earth to obtain a corrected value to be used in the calculation of e/m .
2. Note that you are to measure the radius for the most energetic electrons (the outer edge of the beam). The beam included electrons of a range of energies. We might expect the effective accelerating voltages to range from $V - V_{\text{filament}}$ to V . See if this information can be used to help with error analysis or to explain discrepancies.
3. Using the equation described in the theory section, we can combine variables to get a master equation for the e/m ratio:

$$\frac{e}{m} = \left(\frac{3.91 a^2}{\mu_o^2 N^2} \right) \left(\frac{V}{I^2 r^2} \right).$$

Note that the correct value is 1.7588×10^{11} C/kg.

D	$r=D/2$	V	V_{filament}	I_{coil}	$I_{\text{corrected}}$	e/m

Table. Data for the e/m measurements

5. Analysis and Questions

1. Estimate the uncertainties on your measurements of the directly measured quantities V , I , and r (assume that a is a constant), and from these determine an uncertainty *for every individual measurement* of e/m . Take the average of all of measurements of e/m values and determine the standard deviation. Compare the average of the individual uncertainties with the standard deviation.
2. Compare your results for e/m with the accepted value. Do they agree within experimental uncertainties?
3. How high an accelerating voltage would be required before relativistic effects increase the observed mass by 1 %?